

Imperial Business School

Applied Trading Strategies (BUSI97075-2526)

For this project, each group will have to develop a trend-following strategy in commodities.

This assignment is designed to assess your ability to formulate sensible investment ideas, implement them logically, and then evaluate them rigorously.

This is not a pure forecasting exercise. Nor is it competition to achieve the highest out-of-sample Sharpe ratio. A strong project combines clear economic or statistical reasoning, sound backtesting methodology, thoughtful portfolio construction, and an honest discussion of the strategy's strengths and weaknesses.

Data provided

You will be given two daily frequency datasets:

The first dataset contains daily excess return index data for **23 commodities futures** (sheet titled "Return Indices"), and the sectors to which these commodities belong (sheet titled "Assets").

This defines the investable universe, and these are the only instruments you can trade.

The second dataset contains **auxiliary market variables** to be used (in Task 2 and Task 3) for strategy enhancements. These variables include:

- foreign exchange spot rates
- ATM implied volatilities
- short-rate and yield information, including 3-month deposit rates and 10-year yields
- equity indices
- credit spreads
- short-dated forward rates such as the 1-month EUR forward

Train and test split:

The dataset should be divided into a training set and a test set. As a guideline, use data up to 31 December 2019 as the training period and data from 1 January 2020 onwards as the test period.

This is one of the most important aspects of the project. **You must avoid look-ahead bias at all costs.**

All strategy design choices, model specification decisions, and hyperparameter selection must be carried out exclusively within the training sample. The training set may in turn be subdivided into training and validation periods, for example through rolling or forward cross-validation.

If you are unsure whether a particular choice may introduce look-ahead bias, your group **MUST** discuss this with Francesco, the TA, during lab sessions.

Task 0 – EDA

Begin with an exploratory analysis of the commodities dataset, including summary statistics for the asset returns for each series - # of observations, mean, standard deviation, correlations, etc.

Next, construct a **commodity market factor**, defined as the equally weighted average return of all commodities in the dataset.

Task 1 – Ideas First: Momentum / Trend Strategy Design

Using an “ideas-first” approach, develop a time-series momentum / trend-following strategy for each commodity. The signal should be based on past returns of its own time series (or related technical/economic signals for that asset) but **not on a cross-section basis**.

You are free to choose the lookback and rebalancing frequency and any portfolio construction / weighting method you want, but you should justify your choice clearly. If you develop multiple trend signals, you may combine them into an ensemble strategy.

For example, you may build your strategy using one of the types of signals discussed in class:

- Historical 12-month returns (e.g. Moskowitz, Ooi, Pedersen, JFE 2012)/
- Adaptive TSMOM
- Moving-average crossovers
- Lookback straddle-type signals
- Gaussian-mixture-based signals

Or your group can try your own trend-following strategy!

- First, calculate the summary statistics for the strategy as well as for the commodity market factor calculated in Task 0.
- Next, show a plot comparing the cumulative strategy performance.
- Explain which parameters you are using in your backtest and why. Did you try other parameters/strategies before you picked your final set?
- Run a regression of your strategy returns on the commodity market factor to calculate the sensitivity with respect to the commodity market.
- Show the contribution of each sector (“Agri & Livestock”, “Energy”, and “Metals”) to the total strategy performance.

Task 2 – Strategy Enhancements

In this part, you should improve on the strategy developed in the previous task through a limited number of well-motivated enhancements. Depth is more important than breadth. The objective is not to overcomplicate the strategy, but to choose a few meaningful extensions and study them carefully. They must be non-data driven.

Examples:

- Signal construction – enhance by adding counter-trend signals (breakout, RSI, pairs trading) using all commodities or other economically motivated extensions
- Portfolio construction - extend techniques used in Task 1 aggregation - e.g., mean-variance optimisation, risk parity or another allocation method you deem appropriate – justify your choice.
- Improved risk management – drawdown control, risk budgeting, etc.
- Transaction cost analysis: include proportional transaction costs of say 0.25% and study whether the strategy remains robust; Consider turnover/liquidity constraints.

You are strongly encouraged to implement the enhancement suggestion below – YOU WILL BE REWARDED.

- Design an ideas-first enhancement based on economic reasoning rather than pure pattern search. Here, you must use at least one of the auxiliary market variables from the second dataset. Or you can **find your own data and explain the rationale**. Note: the only tradable assets remain the 23 commodities in the first dataset – you are still predicting forward returns after adding new information.

Example of possible improvements include:

- cross-market relationships (between DXY and precious metal commodities for e.g.)
- volatility-based filtering (based on equities or bonds as in VIX/MOVE)
- use of credit, equity, FX, or rates information as a regime indicator

Task 3 – Data-driven strategies

Build at least one data-driven strategy that attempts to extract predictive signals using statistical or machine-learning methods.

Admissible methods range from simple linear models, such as linear, ridge, or lasso regression, to more flexible approaches including tree-based methods or neural networks.

The objective is NOT to maximise model complexity. A simple model that is well specified, interpretable, and robust is preferable to a complex but unstable one.

You must discuss feature design, target choice, model training, validation, and portfolio construction.

You need to use the auxiliary features in the second dataset for this ML part but may also include additional features or technical indicators of your own, provided the full pipeline is free from look-ahead bias.

Deliverables

You will be assessed on both the code and the report, with greater weight placed on the report.

Your report should be professional, concise, and analytical. The 4 tasks delineated above should be sectioned – your report should be clear on the methodology, assumptions, results (tables/graphs), possible enhancements and challenges. Make sure to leave sufficient space for the conclusion, including lessons learned and any findings that may be relevant for the live trading of your strategy.

Maximum length: 10 pages.

Coding suggestions:

It is strongly recommended that strategy modules/functions return outputs in a standardised format, such as a pandas.DataFrame or numpy array, where each row contains the signals for the commodity universe at a given date.

Similarly, portfolio construction functions or classes should return outputs in a consistent format, for example as a pandas.DataFrame or numpy array, where each row contains the portfolio weights at a given date.

This type of standardisation makes it much easier to compare strategies, swap allocation methods, and maintain a clean backtesting pipeline

Please provide a one-page summary of all classes/methods/functions used and a clear explanation of how the code has been organised. Your code must run in a Windows or a MAC environment. If there are any problems with your code, explain where it breaks etc. in the one-page summary. (Without any disclaimers, the assumption is that everything works from start to finish.)

Francesco will provide further details during the lab sessions/Python tutorials and explain the type of code expected.

Evaluation criteria

- **Clarity of Design:** clear motivation, structure, and rationale behind the strategies.
- **Statistical Evaluation:** sound use of performance measures, validation procedures, and empirical tests.
- **Depth of Analysis:** ability to explore at least one aspect of the project in genuine depth.
- **Code Quality:** readability, structure, documentation, and reproducibility.
- **Initiative and Judgment:** evidence of original thinking, sensible simplifications, and disciplined choices.

Final note

The strongest projects will combine economic intuition, careful implementation, and honest empirical evaluation. Your goal is not to produce the most computationally intensive project, but to demonstrate that your choices are informed, thoughtful, and well justified.

Large Language Models may be used as supporting tools. However, over-reliance on AI to submit generic or superficial ideas can be detected easily and will be penalised.

We value insight over brute force. Avoid unnecessary grid searches / testing hundreds of parameter combinations without a valid reason.

Good Luck Backtesting!